

# CS Core Subjects – Quick Notes

## DBMS

- DBMS manages and organizes data in databases.
- Key concepts: SQL, primary key, foreign key, normalization, indexing, transactions.
- ACID: Atomicity, Consistency, Isolation, Durability.

## Operating Systems

- OS manages hardware resources and provides services to applications.
- Key concepts: processes, threads, CPU scheduling, deadlock, memory management, virtual memory.
- Common scheduling algorithms: FCFS, SJF, Round Robin, Priority.

## Computer Networks

- A computer network enables devices to communicate and share resources.
- OSI layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.
- TCP is connection-oriented and reliable; UDP is connectionless and faster.

## OOP

- Core principles: Encapsulation, Abstraction, Inheritance, Polymorphism.
- Class is a blueprint; object is an instance of a class.
- Method overloading is compile-time polymorphism; overriding is runtime polymorphism.

## Data Structures

- Common structures: arrays, linked lists, stacks, queues, trees, graphs, hash tables.
- Stack follows LIFO; queue follows FIFO.
- Binary search on a sorted array takes  $O(\log n)$  time.

*Purpose: Small sample PDF for testing document/API upload and Postman connections.*